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Understanding Phase and Microstructure Evolution of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene-Polymer Derived Silicon Carbide

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Abstract

Incorporating MXenes as additives into polymeric ceramic precursors allows homogenous integration of these 2D carbides into ceramic matrices. In this study, $\text{Ti}_3\text{C}_2\text{T}_x$ MXene flakes were functionalized using 3-(2-aminoethylamino)propyltrimethoxysilane (AEAPTMS) and stabilized in allylhydrido-polycarbosilane (SMP-10), a SiC polymeric precursor. Thermogravimetric analysis suggests the composites retain more residual mass (~2.45-4.21% mass loss) in the 1200-1400 °C range, indicating enhanced ceramic yield for 2 wt.% MXene addition. In-situ high temperature X-ray diffraction on SMP-10-derived SiC for the first time indicates faster but suppressed crystallization initiation in MXene-incorporated SiC. Microstructural characterizations indicate formation of a <20 nm thick Ti-Si-

C interfacial nanostructure between distinct SiC and MXene-derived TiC_y phases, with greater defect accumulations at the grain interfaces. These findings are the first step towards realizing the potential of 2D MXenes as additives in polymer-derived ceramics (PDCs) for advanced engineering applications.

Keywords: MXene; polymer-derived ceramics; in situ high-temperature XRD; interfacial nanostructure; SiC composites

1. Introduction

The family of two-dimensional transition metal carbides (MXenes), discovered in 2011, represents the 2D analogues of bulk transition metal carbides and has rapidly emerged as versatile functional materials and additives.^{1, 2, 3, 4, 5, 6} Their applications span diverse fields such as energy storage, sensors, catalysis, bioengineered materials, and, more recently, as fundamental building blocks for developing materials suitable for extreme environmental conditions.^{7, 8, 9} MXenes are atomically thin (~ 1 nm), solution-processable, carbide/carbonitride materials characterized by the general chemical formula of $\text{M}_{n+1}\text{X}_n\text{T}_x$. Here, M denotes a transition metal from groups 3-6 of the periodic table; X refers to carbon and/or nitrogen atoms interleaved between layers of M; and T_x represents various surface functional groups introduced during the top-down chemical exfoliation process from their corresponding MAX phase precursors.¹⁰

Using MXenes as 2D additives in polymer-derived ceramics (PDCs) represents an exciting area for materials under extreme environments.^{11, 12} Ceramics synthesized via polymeric routes represent a significant advancement in the fabrication of near-net-shape advanced materials, owing to their combination of thermal stability and processing versatility.¹³ These materials are produced through the pyrolysis of preceramic polymers such as polycarbosilanes, polysiloxanes, and polysilazanes under inert atmospheres while enabling precise control over the composition, phase distribution, and microstructural characteristics of the resulting ceramic materials.¹⁴ MXenes, characterized by their exceptional electrical conductivity, catalytic performance, mechanical strength, and high-temperature stability, can enhance the intrinsic properties of PDCs, such as their electrical and thermal transport behaviors, oxidation resistance, thermal stability, and mechanical robustness, leveraging the combination of properties from both material classes. Such hybrid composites could potentially exhibit tailored multifunctionality suitable for demanding high-temperature environments.

While MXenes have been recently incorporated into PDCs, particularly $\text{Ti}_3\text{C}_2\text{T}_x$ in the SiOC system, a fundamental understanding of their integration mechanisms and thermal behaviors in oxide-free ceramics is yet to be established.¹⁵ Further, these oxide-containing matrices promote the formation of Ti-oxides and glassy interphases, complicating interpretation of carbide-carbide interactions. In contrast, SMP-10-derived SiC provides an essentially oxide-free, carbide-rich environment that is chemically closer to $\text{Ti}_3\text{C}_2/\text{TiC}$, enabling us to isolate and study carbide-carbide interfacial

phenomena and MXene phase evolution without the confounding effects of an SiO_x network. In addition, unlike graphene or hexagonal boron nitride (*h*-BN) reinforcements, which typically act as chemically passive fillers, $\text{Ti}_3\text{C}_2\text{T}_x$ MXene actively participates in the ceramization process. Its ability to undergo a controlled 2D-to-3D phase transformation at high temperatures facilitates the formation of semi-coherent carbide interfaces, offering a distinct interfacial interaction mechanism compared to the weak interfacial interactions often observed in graphene-SiC systems. Surface chemistry of MXenes and the incompatibilities with PDCs lead to inhomogeneous mixing and phase separation in the pre-ceramic polymer matrix. Additionally, interfacial bonding mechanisms between MXene functional groups and evolving ceramic phases during polymer-to-ceramic conversion lack systematic characterization, particularly regarding how transformations and adsorption kinetics influence the final composite architecture and the grain interfaces in sintered products.

This study reports a single-step design strategy to functionalize inherently hydrophilic MXenes and then incorporate them homogeneously into preceramic polymers using $\text{Ti}_3\text{C}_2\text{T}_x$ MXene as the 2D functional additive, and allyl hydrido polycarbosilane (SMP-10) as the silicon carbide (SiC) polymer precursor. A central question we address is how the interface interactions between the MXene and the polymer matrix evolve during thermally activated phase transformation and pyrolysis, how this interfacial behavior impacts the resulting microstructure and the stability of the Ti_3C_2 -SiC ceramic composites. To this end, we first show that surface

functionalization of MXene flakes via organic silane coupling agents enhances their compatibility with the non-polar ceramic precursor using spectroscopy tools (FTIR and XPS) and X-ray diffraction (XRD). We further evaluate the functionalized MXene-SMP10's stability during and post cross-linking stage and early pyrolysis using thermogravimetric analysis (TGA) and scanning electron microscopy (SEM). After establishing phase stability through the initial processing steps, we discuss the phase transformation of the MXene-SMP-10 composites using in-situ hot-stage XRD up to 1600 °C. Further, we investigate the high-temperature behaviors of the composites, focusing on phase evolution and interfacial transformations using scanning transmission electron microscopy (STEM) equipped with energy dispersive X-ray spectroscopy (STEM-EDS) and atomic resolution STEM high-angle annular dark-field (HAADF) imaging. This approach enables us to directly correlate thermal transformation with the evolving microstructure and allows us to map the performance of the resulting $\text{Ti}_3\text{C}_2\text{-SiC}$ ceramics in future studies.

Materials and Methods

Synthesis of MAX phase precursor Ti_3AlC_2

Ti_3AlC_2 MAX phase precursor was prepared through reactive sintering of elemental powders combined in a stoichiometric ratio of TiC:Al:Ti (2:1:1). For a 50 g batch, 30.77 g TiC, 6.93 g Al, and 12.3 g Ti powders were used. Following established protocols, the elemental powders were homogenized in a tumbler ball mill with zirconia grinding media for 30 min, though no

significant particle size reduction occurred during this process. The homogenized powder mixture was then transferred to an alumina crucible, covered with a graphite foil lid, and pressurelessly sintered inside a tube furnace under atmospheric pressure with continuous argon flow. Optimal sintering parameters included a temperature of 1400°C with a 4 h isothermal hold, controlled heating rate of 1.5 °C/min, and cooling rate of 1.5 °C/min. Post-synthesis, the sintered MAX phase material was milled and sieved through a 70 µm mesh to obtain a powder with particle sizes ranging from 20-70 µm for subsequent processing steps.

Synthesis of MXene ($Ti_3C_2T_x$)

$Ti_3C_2T_x$ MXene was subsequently obtained by aqueous solution-based selective etching of Ti_3AlC_2 powders via the MILD method using lithium fluoride (LiF, ACS reagent, Sigma Aldrich, 99.9%) with hydrochloric acid (HCl, 36-38 wt.%, ACS reagent, Sigma Aldrich, 99.9%) as etchants. In a typical synthesis experiment, 0.5 g Ti_3AlC_2 powders were slowly added to a solution of LiF (~2 g) and HCl (10 ml), and then stirred for 48 h at 50 °C in an oil bath. Thereafter, the resulting suspension was diluted by deionized water and transferred to a 50 mL centrifuge tube. The contents were neutralized via successive centrifugation cycles at 4000 rpm for 5 min, until the pH was ~6.5. Further, the etched MXene powders were delaminated using ultrasonication for 1 h and lyophilized at -50 °C for two days. Washing was implemented by multiple rounds of deionized water and diluted HCl (2 mol/L) to remove F^+ and Li^+ cations until the supernatant pH value was

approximately neutral, before being finally washed by ethanol twice. The intercalated $\text{Ti}_3\text{C}_2\text{T}_x$ clay was dispersed in deionized water and centrifuged at 3500 rpm for 1 h to separate the unexfoliated layers. The obtained $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheet dispersion was frozen by liquid nitrogen and freeze-dried at $-50\text{ }^\circ\text{C}$ for 48 h.

$\text{Ti}_3\text{C}_2\text{T}_x$ MXene functionalization

In order to achieve uniform mixing of MXene and polymer precursors, the obtained $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheets were functionalized with N-(2-aminoethyl)-3-aminopropyltrimethoxysilane (AEAPTMS, Sigma-Aldrich) to form a covalently bonded silane coupled hydrophobic surface on the $\text{Ti}_3\text{C}_2\text{T}_x$. Surface functionalization of $\text{Ti}_3\text{C}_2\text{T}_x$ was performed in a water/ethanol mixture (10/90 wt.%). In a typical experiment, 1.5 mg/ml suspension of $\text{Ti}_3\text{C}_2\text{T}_x$ + water/ethanol mixture was stirred at room temperature, and glacial acetic acid (CH_3COOH) was slowly added until the solution reached a pH between 4-5. To ensure sufficient reaction between the -OH terminations of the hydrolyzed $\text{Ti}_3\text{C}_2\text{T}_x$ and the silanol groups of AEAPTMS, the mass ratio of AEAPTMS to $\text{Ti}_3\text{C}_2\text{T}_x$ was set to 1:2. The AEAPTMS/ethanol solution was added dropwise to the $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheet suspensions and stirred for 5 h. Note that the AEAPTMS reagent was added slowly at a rate of 100 $\mu\text{l}/\text{min}$ over 30 minutes to 1 h while the mixtures were continuously stirring. The mixtures were stirred for 7 h after complete addition of the AEAPTMS solution to the system. Subsequently, the suspension was washed four to five times with ethanol by centrifugation at 10,000 rpm until the pH reached ~ 7 ,

and the dispersions were observed to be stable and suspended evenly in the ethanol solution. The obtained AEAPTMS-functionalized $\text{Ti}_3\text{C}_2\text{T}_x$ nanoflakes (AEAPTMS- $\text{Ti}_3\text{C}_2\text{T}_x$) were then collected by a hydrophobic PVDF filter and vacuum-dried at 40 °C until dry.

$\text{Ti}_3\text{C}_2\text{T}_x$ MXene/SMP-10 synthesis

All mixing was carried out in a glove box. To prepare the MXene-SiC composite precursor, SMP-10 (Starfire Systems, Inc., NY) was first diluted in anhydrous toluene (99.8%, Thermo Scientific Chemicals) to reduce its viscosity and enable homogeneous dispersion of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene nanosheets. In a typical experiment, 5 g of SMP-10 was dissolved in 5 ml of toluene. Subsequently, 2 wt.%, 5 wt.%, or 10 wt.% of AEAPTMS-functionalized $\text{Ti}_3\text{C}_2\text{T}_x$ MXene was added to the SMP-10/toluene solution. The mixture was magnetically stirred for 24 h at room temperature to achieve uniform dispersion of the MXene within the polymer matrix. After dispersion, the toluene solvent was evaporated by heating at 120 °C for 6 h. The resulting dry mixture was then transferred into an alumina crucible and pyrolyzed at 1100 °C for 2 h under argon (Ar) atmosphere, using a heating rate of 1.5 °C/min.

Fourier Transform Infrared Spectroscopy (FTIR)

FTIR measurements were performed on a Bruker Alpha FTIR spectrophotometer, configured with a platinum ATR module. The

transmittance measurements were taken over a wavelength range of 2-16 μm . Data were analyzed using Opus software.

X-ray photoelectron spectroscopy (XPS)

XPS measurements were collected on a VersaProbe 5000 with a spot size of 400 μm at an energy resolution of 0.1 eV. The binding energy scale for charge correction was assigned by adjusting the C 1s peak at 284.8 eV. All XPS spectra were analyzed using the CASA XPS software package.

Spark plasma sintering

Pre-pyrolyzed ceramic powders were consolidated via spark plasma sintering (SPS) using a DSP 507 sinter press (Dr. Fritsch Sondermaschinen GmbH, Germany). The powders were loaded into a graphite die with a nominal diameter of 10 mm and subjected to an uniaxial pressure of 45 MPa. Sintering was conducted at target temperatures ranging from 1200 °C to 1500 °C, with a fixed dwell time of 5 min at peak temperature. The thermal cycle employed a constant heating rate of 80 °C/min under an inert argon atmosphere.

Thermogravimetry

The thermal evolution of the pre-pyrolyzed SiC and SiC-Ti₃C₂ composite ceramics was systematically investigated from ambient temperature to 1500 °C using a thermogravimetric analyzer (TGA, Setaram Setsys Evolution) under high-purity helium flow. Quantification of mass loss during

ceramization and pyrolysis products derived from pre-pyrolyzed SiC ceramics was performed to evaluate phase evolution and microstructural transformations. To ensure statistical robustness and reproducibility, mass loss and dimensional shrinkage measurements were conducted in duplicates, and the resulting data were averaged. The reported error bars and $\pm$ values corresponded to the spread between these duplicate measurements and were intended to indicate experimental scatter rather than statistically inferred confidence intervals. The present study focuses on identifying qualitative and semi-quantitative trends in phase evolution and interfacial behavior as a function of MXene content.

SEM/EDS

The surface morphology of the samples pyrolyzed at 1200 °C was investigated by a scanning electron microscope (SEM, ThermoFisher Apreo HiVac) in combination with an energy-dispersive X-ray spectrometer (EDS, EDX Elite 150 SDD with 127 eV resolution). The spot size was maintained at 0.3-5 nm with accelerating voltages between 5-10 kV. For EDS characterization, the samples were sputtered with Pt-Rd for 30 s.

STEM/EDS

TEM samples were prepared using the in-situ liftout technique in a SEM equipped with a focused ion beam (Thermo Scientific Quanta Dual Beam). Selected area electron diffraction (SAED) and STEM-EDS analysis were performed using a Thermo Scientific Titan Themis 200 microscope

equipped with a Super-X EDS system at the Irradiated Materials Characterization Laboratory in Idaho National Lab (INL). Atomic resolution STEM imaging was performed using a probe and image-corrected Thermo Scientific Spectra 300 TEM at INL's Critical Materials and Energy Systems Innovation Center.

In-situ XRD

In situ X-ray diffraction (XRD) analysis was performed using an Anton Paar HTK2000N high-temperature stage integrated with an Anton Paar XRDynamic 500 diffractometer. A colloidal suspension of low-temperature (1100 °C) pyrolyzed SiC or SiC-MXene fine powder, dispersed in 200-proof ethanol, was drop-cast directly onto a tungsten heating filament. After drying under ambient conditions, the samples were sealed under high vacuum ($\approx 1 \times 10^{-5}$ mbar).

Data acquisition followed a step-and-hold thermal protocol to ensure temperature stability during measurement. The sample was heated at a rate of 20 °C min⁻¹ to discrete target temperatures at 20 °C intervals. Upon reaching each setpoint, the temperature was held constant (isothermal dwell) for the duration of the diffraction scan (15 min). XRD patterns were acquired over a 2θ range of 10°–80° during these isothermal holds. Following acquisition, the temperature was ramped to the next setpoint, and the process was repeated up to 1900 °C. In total, approximately 50 scans were collected between 900 °C and 1900 °C, corresponding to a total experiment duration of approximately 750 min (12.5 h) per run. While data was collected

till 1900 °C, only datasets up to 1600 °C are presented in the discussion section, beyond which signal quality was compromised by the heating strip diminishing the diffraction intensity (**see supplementary information Figure S5**).

Results and Discussion

Grafting of MXenes and preparation of MXene-SMP-10 green bodies

$\text{Ti}_3\text{C}_2\text{T}_x$ MXene exhibits negative zeta potentials in aqueous media, typically ranging from -30 to -45 mV at neutral pH,^{16, 17} which presents several challenges for incorporating MXenes into polymer systems. The strong negative charge promotes excessive hydration and stabilization of MXene sheets in aqueous media, making it difficult to achieve homogeneous mixing with hydrophobic polymers.¹⁸ Additionally, the weak interfacial interactions between negatively charged MXenes and polymeric precursors result in phase separation during the green body synthesis stage, which later translates to agglomeration, causing poor stress transfer, ultimately compromising the mechanical properties of the derived composite.

To overcome these processing challenges, we performed silane grafting on the etched (**Figure S1-a,b**) and delaminated (**Figure S1-c**) MXene using AEAPTMS (3-(2-aminoethylamino)propyltrimethoxysilane) as the silane coupling agent on the etched MXene.^{15, 19} First, we lyophilized the MXenes using freeze drying at -50 °C to remove moisture, and subsequently grafted the dry MXene powders with AEAPTMS dissolved in ethanol to replace the functional moieties as shown in **Figure 1a**. Specifically, the two identified

modes of grafting were silanol group ends leading to covalent bonding and the amine group ends leading to electrostatic interactions as reported earlier.¹⁹ Furthermore, the silane groups in AEAPTMS react with hydroxyl groups on the MXene surface along with steric hindrance provided by the functionalization agent to minimize π - π stacking between adjacent $\text{Ti}_3\text{C}_2\text{T}_x$ nanosheets and prevent aggregation.^{20, 21} The structures of AEAPTMS and SMP-10 and the nature of interactions with MXenes are shown in **Figure S2, Figure 1d**.

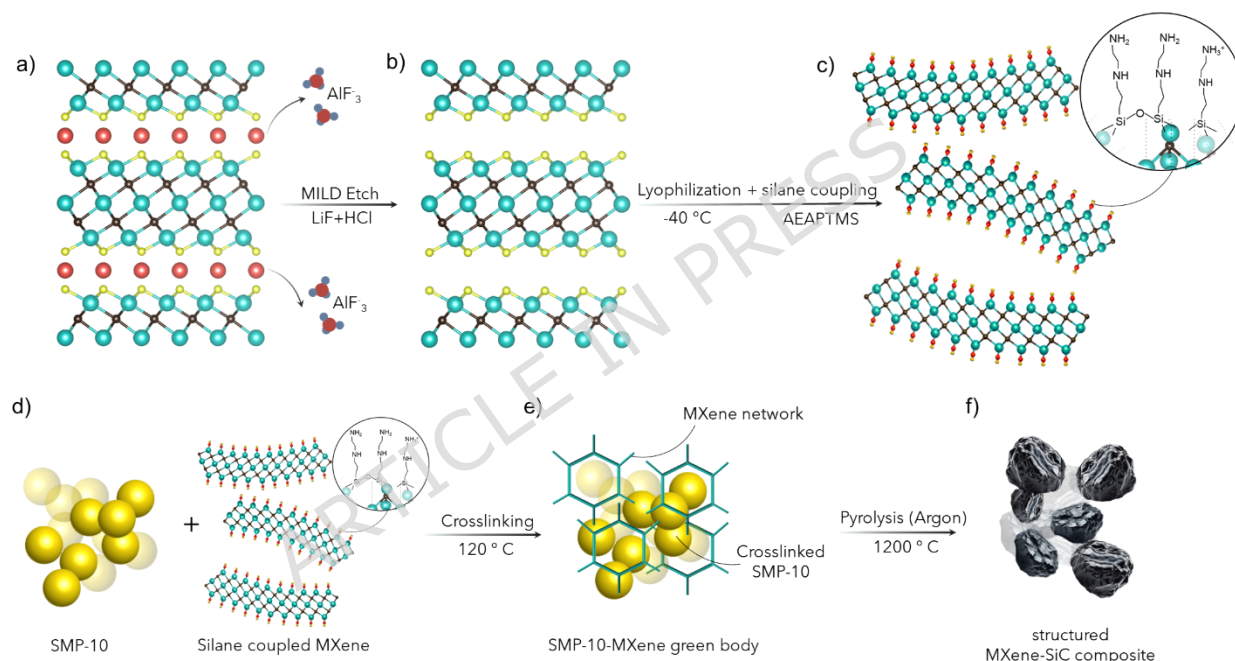


Figure 1: Schematic representation of a) MAX phase to b) MXene synthesis via the MILD etching route, c) delaminated MXenes grafted using AEAPTMS silane coupling agent to enhance MXene stabilities in pre-ceramic polymers, d) synthesis of SMP-10 preceramic polymer and MXene cross-linked green bodies, e) subsequent pyrolysis to develop f) structured MXene-SiC composite powders.

Silane functionalized MXenes were then gradually added to SMP-10 polymeric precursor, and mixed to make slurries of MXene-SMP-10 mixtures in toluene (**Figure 1b**). Crosslinking of the MXene-SMP-10 mixtures was initiated via heating in an inert environment (**Figure 1c**) with subsequent pyrolysis of the green bodies at 1100 °C to obtain pyrolyzed MXene-amorphous SiC composite powders. Pre-pyrolyzed composite green bodies were evaluated using in-situ hot stage XRD to understand crystallization, and then the powders were further pyrolyzed using an electric field-assisted spark plasma process to understand the microstructural evolution.

Compositional and structural evolution

As the first step, we analyzed the FTIR spectra (**Figure 2a**) of different constituting species: pure MXene, MXene functionalized using AEAPTMS, and MXene-SMP-10 blends. In the spectrum for pure MXene, we observe weak -OH stretching between 3600-3200 cm^{-1} and -OH bending modes at 1500-1300 cm^{-1} .²² This Ti-O-H stretching further diminishes after functionalization (shown as a decrease in the hump), indicating the removal of -OH groups and a probable replacement with -Si groups. Since the MXenes were lyophilized before functionalization, the decrease in the hump for functionalized MXene could also be due to the disruption of hydrogen bonding on the surface.¹⁹ We observed an atypical red shift of the Ti-F bending mode at 635 cm^{-1} in the pure MXene, which may be attributed to in-

tandem interactions of $-F$, $=O$, and Ti atoms simultaneously present on the basal planes.²² This peak diminished in the functionalized samples along with other weak $-F$ stretching vibrations between $1300-1000\text{ cm}^{-1}$. An asymmetric vibration mode of the $C-H$ bonds from alkyl chains at $\sim 2850\text{ cm}^{-1}$ was detected in the functionalized MXene, which may be attributed to the effect of interactions with alkyl chains in AEAPTMS during the functionalization process.²¹ Further, a $N-H$ bending vibration was observed at 1578 cm^{-1} in the functionalized samples and attributed to primary amine groups. There was no significant indication of silane functionalization on the MXene as observed with FTIR.

The MXene + SMP-10 precursor FTIR spectrum exhibits characteristic vibrational modes associated with polycarbosilane molecular structures. The fingerprint region ($1100-600\text{ cm}^{-1}$) displays multiple well-resolved absorption bands corresponding to silicon-centered functional groups: $Si-CH_2$ symmetric deformation ($\sim 650\text{ cm}^{-1}$), $Si-C$ stretching vibration ($\sim 800\text{ cm}^{-1}$), $Si-H$ bending ($\sim 900\text{ cm}^{-1}$), and $Si-CH_2-Si$ asymmetric stretching ($\sim 1000\text{ cm}^{-1}$). A sharp, high-intensity absorption band centered at 2120 cm^{-1} is attributed to $Si-H$ stretching vibrations, characteristic of hydrosilane functionalities in the precursor backbone. The spectrum further shows olefinic moieties through the presence of $C=C$ stretching modes at $\sim 1630\text{ cm}^{-1}$ and associated vinyl/allyl $C-H$ stretching vibrations at 3070 cm^{-1} , indicative of unsaturated terminal groups within the polycarbosilane precursor structure.²³

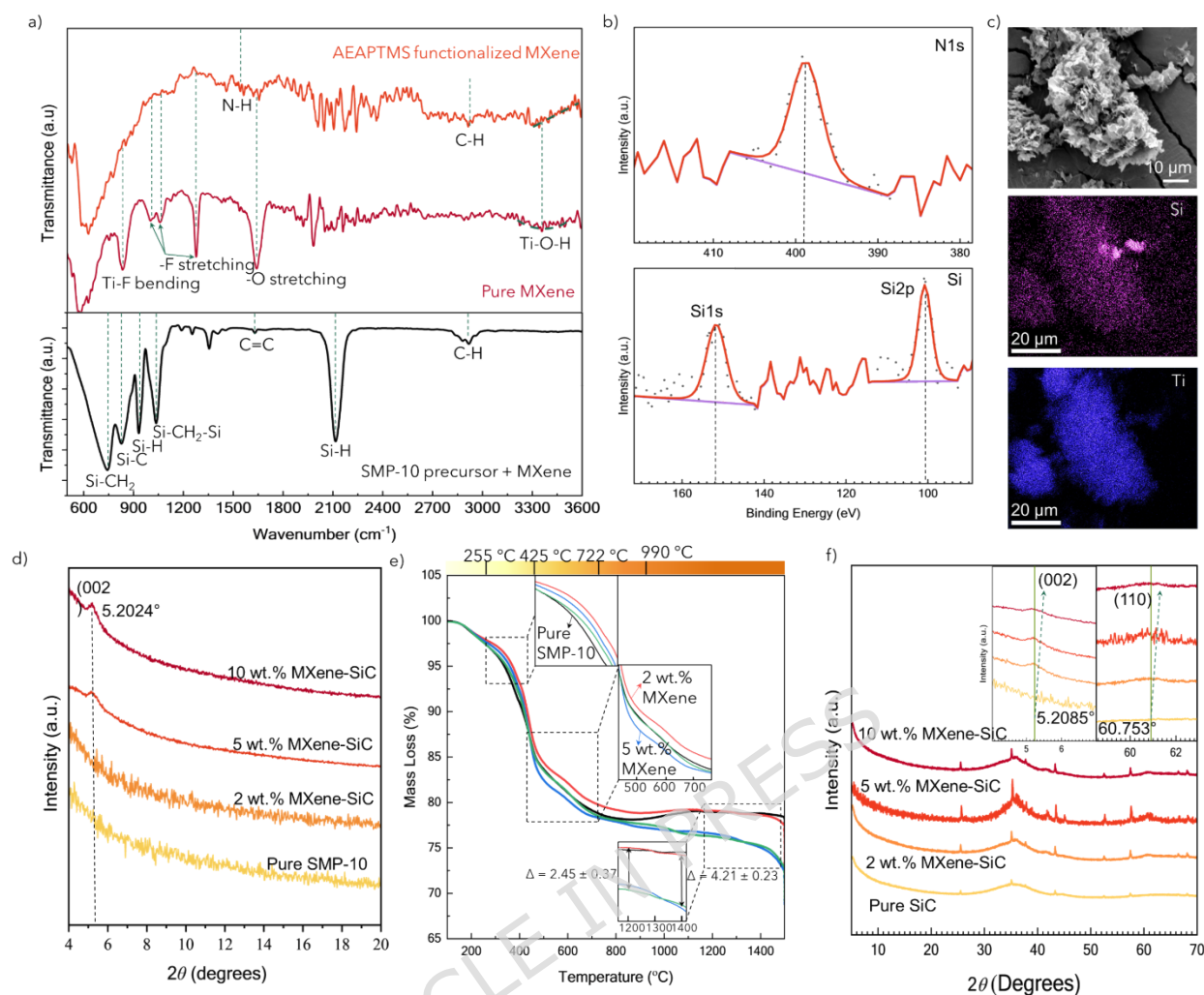


Figure 2: a) FTIR spectra of the pure MXene, functionalized MXene, and SMP-10-MXene (5 wt.%) blends, b) XPS spectra of the functionalized MXene showing Si and N resolved peaks, indicating functionalization, c) EDS maps showing Si and Ti distribution semi-quantitatively and verifying functionalization, d) XRD patterns of the pure SMP-10 with increasing MXene content (2, 5, and 10 wt.%) of the crosslinked samples, e) TGA curves for the pure SMP-10 and with increasing MXene content (2, 5, and 10 wt.%), f) XRD patterns for the pure SMP-10 and with increasing MXene content (2, 5, and 10 wt.%) after pyrolysis at 1100 °C.

The XPS spectrum of pristine $\text{Ti}_3\text{C}_2\text{T}_x$ MXene exhibits an abundance of fluorine terminations,²⁴ a feature previously noted to be detrimental in PDCs, as they have been linked to microcracking and degradation during high-temperature pyrolysis.¹⁵ Upon AEAPTMS functionalization, a marked reduction in the fluorine 1s peak intensity was observed (**Figure S3**), concurrent with the emergence of new spectral features associated with silicon (Si1s at 151.2 eV, Si2p at 100.8 eV) and nitrogen (N1s at 399.2 eV) (**Figure 2b**), corresponding to the nitrogen in the amino groups of the silane coupling agent. These changes show successful covalent grafting of AEAPTMS onto the MXene surface, resulting in partial replacement of halogens (F and Cl) by N-H and Si-Si/Ti moieties (**Table S1**). The estimated stoichiometric ratios of the functional terminations were $\text{Ti}_3\text{C}_2(\text{O}_{0.13}\text{F}_{0.10}\text{Cl}_{0.07}\text{N}_{0.16}\text{Si}_{0.22})$, which indicates substantial incorporation of Si functional groups under optimized reaction conditions. Complementary energy-dispersive X-ray spectroscopy (EDS) verified the presence of silicon, estimated at $\sim 2.7 \pm 0.32$ at.% (**Figure 2c**).

Although FTIR is a useful tool for tracking qualitative changes in surface functional groups, in the present MXene-SMP-10 system it has limited sensitivity to the thin AEAPTMS silane layer. The Si-O-Si and Si-O-Ti vibrational modes expected from the silane lie in the $1100\text{-}800\text{ cm}^{-1}$ region, which is already populated by intense Si-based bands from the polycarbosilane backbone (Si-C stretching, Si-H bending, and Si-CH₂-Si modes). Because SMP-10 constitutes the bulk of the material, the FTIR response in this region is dominated by the polymer matrix, and any

additional silane-related Si contributions are effectively masked within the broad SMP-10 bands. Moreover, ATR-FTIR probes all materials within the optical penetration depth on the order of micrometers. In contrast, XPS and SEM-EDS probe the near-surface region and directly detect the emergence of -Si and -N signals on the MXene after AEAPTMS treatment, along with a reduction in -F terminations.

XRD has been shown to be effective for detecting minimal MXene loadings in metal/ceramic composites earlier.^{8, 25} The amorphous nature of the SMP-10 precursor further provides an ideal background for isolating MXene-specific reflections. We analyzed the evolution of the characteristic low-angle out-of-plane (002) diffraction peak of MXene across varying MXene concentrations (2, 5, and 10 wt.%), with the diffractograms showing a progressive intensification of the (002) reflection centered at $2\theta = 5.24^\circ$ with increasing MXene content (**Figure 2d**). As expected, the 2 wt.% sample exhibited no detectable out-of-plane reflections due to the low concentration, while becoming distinctly visible in the 5 and 10 wt.% composites. This peak position corresponds to an interlayer d-spacing of 1.69 nm, typical of delaminated and functionalized MXene flakes.

Pyrolysis behaviors

The thermal transformation pathway exhibits four distinct transformation regimes (**Figure 2e**), correlating with the temperature markers at 255 °C, 425 °C, 722°C, and 990 °C. Initially, all samples show relative thermal stability up to approximately 255 °C with negligible mass

loss. The first decomposition stage occurs between 255-425 °C, where all compositions lose approximately 5-7% mass, likely corresponding to the evolution of low molecular weight oligomers and initiation of cross-linking reactions in the polycarbosilane structure.²³ The second and most pronounced decomposition phase (425-722°C) exhibits a steep mass reduction of 10-13%, representing the primary polymer-to-ceramic conversion process where hydrogen elimination, Si-H bond scission, and initial amorphous SiC network formation occur. Between 722-990°C, the decomposition rate decreases considerably as the samples approach the ceramic state, with mass loss curves beginning to differentiate based on the MXene content. Typically, MXenes also show accelerated desorption kinetics of the functional groups in this temperature regime.^{26, 27} The higher temperature regime (990-1500°C) shows a composition-dependent behavior, with the 2 wt.% MXene composite (red curve) consistently maintaining higher residual mass compared to both pure SMP-10 (black) and 5 and 10 wt.% MXene samples (blue/green). Within the limited set of loadings examined (0, 2, 5, and 10 wt.% MXene), the 2 wt.% MXene blend exhibits the highest residual mass in this temperature range, suggesting that an intermediate MXene content in this regime can more effectively inhibit mass transport during structural reorganization and promote ceramic network stabilization under our processing conditions, which we view as a compositional trend within the investigated window.

The magnified insets show the composition-dependent mass loss in the 1200-1400 °C regime, showing mass differentials of $2.45 \pm 0.37\%$ (at 1200 °C)

and $4.21 \pm 0.23\%$ (at 1400 °C) between the 2 wt.% and 5 wt.% MXene compositions. The positive differential indicates that the 5 wt.% sample loses more mass than the 2 wt.% sample in this temperature range. The enhanced mass loss observed in the samples with higher MXene loadings (5 and 10 wt.%) at temperatures above 1000°C can be attributed to several interconnected mechanisms: At high temperatures, MXene sheets undergo complete phase transformation to non-stoichiometric titanium carbide (TiC_y), restructuring from a 2D layered configuration to a 3D cubic lattice.²⁸ This transformation process is concentration-dependent, with higher MXene loadings (5 wt.%) experiencing more extensive mass transport during reorganization. While the 2 wt.% loading appears to represent a threshold where MXene sheets effectively stabilize the evolving ceramic network, the 5 wt.% and 10 wt.% concentrations likely alter this balance. In earlier studies, we also observed the evolution of carbon in the form of CO during thermal treatment of MXenes beyond 1000 °C, which indicates the formation of gaseous phases during the process.²⁹ This evolution is however, limited by the amount of oxygen present in the system. Further, the aminosilane coupling agent typically follows similar decomposition patterns to the other functional groups in MXene in this regime and leads to greater mass loss for 5 wt.% and 10 wt.% samples, due to greater vol.% of these terminal groups on MXenes with increased loadings in the matrix.

The presence of MXene appears to hinder mass transport and suppress excessive volatilization, supporting better structural integrity during pyrolysis by acting as structural anchors during polymer-to-ceramic

transformation. The DTA plots (**Figure S4**) show the endothermic peak indicating onset and endset of the primary decomposition of the backbone at ~ 940 °C compared to samples with MXene, which exhibit a narrower window (925-935 °C), indicating a more controlled decomposition with MXene addition. The presence of MXene up to 2 wt.% likely represents a balance where MXene promotes conversion without introducing instability, as we observed greater mass loss in the samples containing 5 and 10 wt.% MXene compared to the 2 wt.% MXene sample at 1200 °C ($\Delta = 2.45 \pm 0.37$). The corresponding DTA plots are given in **Figure S4** with the exothermic events discussed in the supplementary text **section S4**.

Further, we analyzed the evolution of these SMP-10-derived SiC ceramics with varying MXene loadings (0, 2, 5, and 10 wt.%) pyrolyzed at 1100 °C. The XRD plots (**Figure 2f**) show that all samples display broad diffraction peaks in the $30\text{--}40^\circ$ 2θ range, consistent with the amorphous regimes observed at lower temperatures. At 1100 °C, SMP-10 precursor begins its transformation to β -SiC, though complete crystallization typically requires higher temperatures ($\sim 1300^\circ\text{C}$).¹³ A distinct low-angle peak at $2\theta = 5.2085^\circ$ corresponds to the characteristic (002) basal plane reflection of $\text{Ti}_3\text{C}_2\text{T}_x$ MXene in the ex-situ XRD patterns of bulk powder samples, indicating retention of the layered MXene structure after pyrolysis at 1100 °C. **Figure 2f-inset** highlights the in-plane (110) peak of MXene at $2\theta = 60.753^\circ$ for different MXene content samples, which becomes increasingly pronounced with higher MXene content.

In-situ high temperature XRD analysis

To further understand the crystallization behavior and phase evolution of MXene-SiC composites at elevated temperatures, we conducted in-situ high-temperature XRD analyses on pre-pyrolyzed samples up to 1600 °C. This approach enables direct observation of structural transformations and possible phase reactions as a continuous function of temperature.

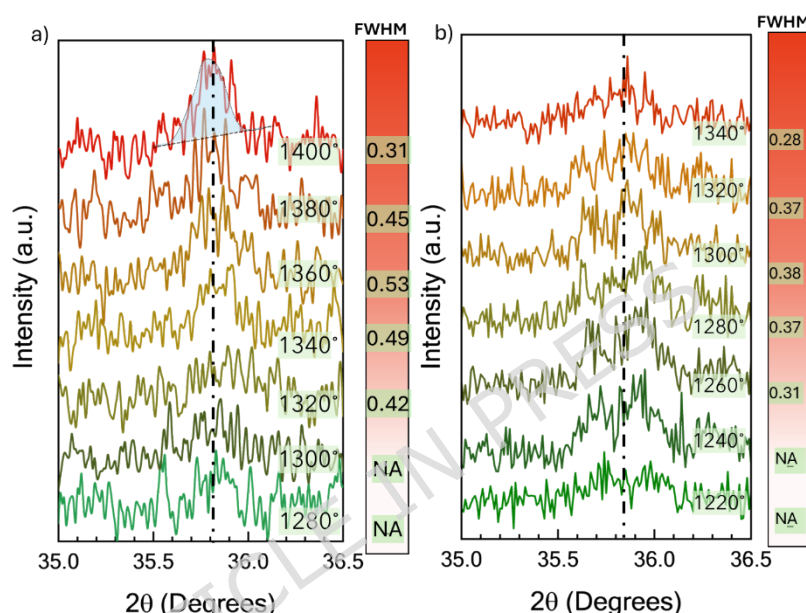


Figure 3: XRD patterns of a) pure SiC and b) $\text{Ti}_3\text{C}_2\text{T}_x$ (5 wt.%) - SiC between temperature ranges 1200° to 1400 °C acquired via in-situ X-ray diffraction. The initiation of crystallization occurs between 1320-1340 °C for the pure SiC, while there is an early initiation from 1240 °C onwards but simultaneous suppression of overall crystallinity in the MXene-SiC samples.

In the pure SiC samples (**Figure 3a**), the emergence of the (111) diffraction peak at $2\theta \approx 35.7^\circ$ is first observable at 1320 °C, becoming progressively more intense at 1340 °C and 1360 °C. The full width at half

maximum (FWHM), which is quantifiable from 1320 °C onward, exhibits a marked decrease with increasing temperature. This FWHM reduction (from 0.42 at 1320 °C to 0.31 at 1400 °C) signifies crystallite growth and improved crystallinity, suggesting that the transformation from amorphous to crystalline SiC initiates in the 1320-1340 °C window and proceeds with enhanced crystallization beyond 1340 °C. The most pronounced sharpening of the peak occurs between 1340 °C and 1360 °C, consistent with bulk crystallization and growth of β -SiC nanocrystallites.³⁰

In contrast, the MXene-SiC composite samples (**Figure 3b**) show an earlier onset of the β -SiC (111) peak, first appearing at 1240 °C and evolving through to 1340 °C. However, the FWHMs of the (111) peak in the MXene-SiC system remain relatively broad and show minimal sharpening across the temperature range, indicating limited crystallite growth and poor coalescence despite progressive nucleation. This deviation in crystallization behavior compared to pure SiC indicates that MXene addition lowers the apparent onset temperature for detectable β -SiC formation, while the reduced degree of peak narrowing with temperature suggests that the combined effects of crystallite growth and strain relaxation are less pronounced in the composite. The in-situ XRD data are consistent with earlier nucleation accompanied by limited coarsening of the β -SiC crystallites in the presence of MXene. Because the FWHM still reflects contributions from both crystallite size and microstrain/defects, we consider this a qualitative indication rather than a quantitative measurement of growth kinetics. One plausible explanation is that the MXene-derived carbide and its preferential

growth along the (111) planes are due to its layered structure,²⁸ hinder long-range atomic diffusion, or introduce interfacial strain, thereby restricting the extent of crystallite growth even as nucleation is facilitated.¹⁰ Additionally, MXene inherently contains Ti and C vacancies, which can create localized energy wells that trap diffusing atoms, thereby reducing their mobility and preventing efficient mass transport, which is mandated for crystallite growth.

We note that reliable Scherrer or Williamson-Hall analysis is not feasible on these in-situ patterns due to thermal diffuse scattering, low sample mass, and tungsten substrate contributions at high temperatures: factors that introduce errors into peak profile fitting. The FWHM trend is therefore interpreted qualitatively as consistent with limited crystallite coarsening in the presence of MXene, rather than as a quantitative grain size measurement. This interpretation is independently supported by HRTEM observation of a high stacking fault density and SAED-confirmed semi-coherence later in the discussion section.

We note that the in-situ XRD experiments were not able to resolve MXene-specific peaks due to a narrow detector window that hindered detection for small specimen masses unlike the ex-situ bulk powder measurements (**Figures 2d, 2f**). The characteristic in-plane (110) diffraction peak of MXenes, typically observed near 60.6° , was also obscured, likely due to overlap with a strong tungsten peak centered around 60° , originating from the heating element. Additionally, diffraction signals corresponding to WO_3 were detected beyond 1300°C , which we attribute to the partial oxidation of

the tungsten heating element during high-temperature annealing (**Figure S5**).

The in-situ XRD data for the pure SiC and MXene-SiC composites indicate two trends. First, the (111) reflection of β -SiC in the composite appears at a lower temperature (1240 °C) than in the MXene-free SiC (1320 °C), indicating an earlier onset of detectable crystallization. Second, whereas the (111) peak in the pure SiC progressively sharpens between 1320 °C and 1400 °C, the corresponding peak in the MXene-containing samples remains comparatively broad over the same temperature range, with only modest FWHM reduction. These trends point to earlier nucleation but limited crystallite coarsening in the presence of MXene.

Due to the narrow detector window, overlap of MXene reflections with tungsten peaks, and the emergence of WO_x peaks from partial oxidation of the heating element at high temperatures, MXene-specific phases cannot be directly resolved in the in-situ XRD patterns. Consequently, the underlying microscopic mechanisms cannot be established from in-situ XRD alone. Instead, we interpret the combination of (i) earlier β -SiC nucleation and (ii) sustained peak broadening as consistent with MXenes acting as crystallization modifiers: they likely provide heterogeneous nucleation sites while simultaneously introducing interfacial strain and diffusion barriers that limit grain growth. This interpretation is supported by ex-situ analyses of sintered samples, which show MXene-derived TiC_y at SiC grain boundaries, a Ti-rich interfacial nanolayer, and a high density of stacking faults near the interface which we will discuss in the following sections.

Understanding of sintered MXene-SMP-10 samples

To evaluate the microstructural changes due to MXene incorporation under densification conditions, the pyrolyzed MXene-SMP-10 powders were consolidated via spark plasma sintering (SPS) at 1500 °C under uniaxial pressures of 45 MPa (**Figure 4a**). The relative densities of the sintered samples were between 82-87% (**Table S2**). XRD analysis of the sintered samples suggests the formation of β -SiC as the primary phase, with minor presence of α -SiC (**Figure 4b**).

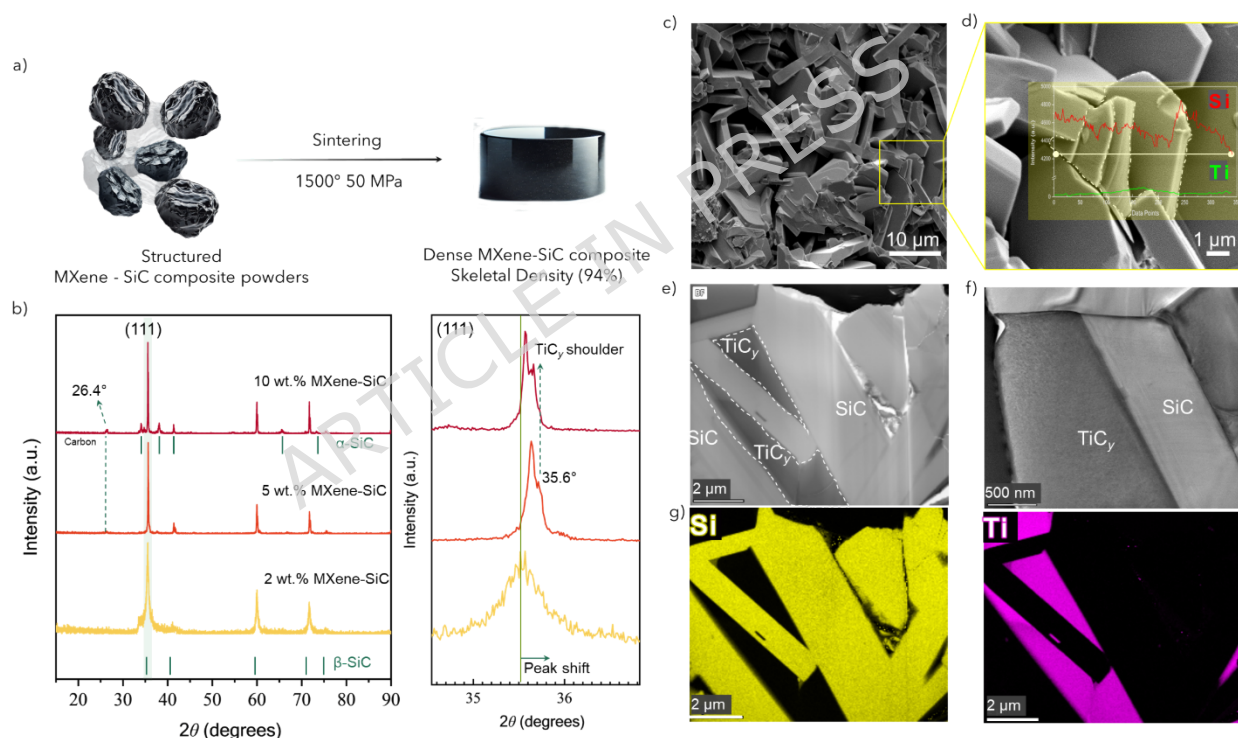


Figure 4: a) Schematic showing consolidation of pyrolyzed MXene-SMP-10 powders using SPS, b) XRD patterns of composites with increasing MXene content (2, 5, and 10 wt.%) after sintering, (111) peak shifts with an increase in MXene loadings, c,d) SEM micrographs of the sintered 5 wt.% MXene

content sample showing texture evolution and line EDS showing Si and Ti presence at grain interfaces, e,f) site-specific TEM micrographs showing SiC and Ti-rich regions with sharp contrasts in FIBed samples, g) corresponding EDS maps showing distinct Si and Ti regions, carbon EDS is not presented due to larger errors in quantifying carbon using EDS.

The dominance of the cubic polymorph is consistent with the known kinetic favorability of this phase under the rapid heating and densification conditions of SPS.³¹ With incremental addition of MXene (2, 5, and 10 wt.%), a rightward shift of the (111) peak of β -SiC ($2\theta \approx 35.56^\circ$) is observed (**Figure 4b-right panel**). This rightward shift corresponds to a reduction in the interplanar spacing of the (111) plane, indicating lattice contraction within the SiC structure, likely induced by strain fields at the SiC-TiC_y interfaces or the incorporation of residual species/defects introduced during MXene phase transformation. A prominent emergence of a distinct shoulder on the (111) SiC peak at $2\theta \approx 35.67^\circ$ shows formation of non-stoichiometric TiC_y at the interface, leading to some strain development.

The presence of a MXene-SiC interface introduced compressive residual lattice strain in the range of 0.39% to 0.53% in the MXene-containing samples (**see Supplementary Text S5**). These finite lattice misfits, together with known CTE differences, indicate the presence of local residual stress fields at the TiC_y-SiC interfaces. Upon cooling from the sintering temperature, the secondary phases (TiC_y and α -SiC) with higher coefficients of thermal expansion ($\sim 7-8 \times 10^{-6} \text{ K}^{-1}$ and $4.7 \times 10^{-6} \text{ K}^{-1}$, respectively) contract more

than the β -SiC matrix ($3.8 \times 10^{-6} \text{ K}^{-1}$).^{32, 33} Under the constraint of a semi-coherent interface, this differential contraction exerts a compressive stress on the β -SiC matrix, consistent with the observed reduction in d-spacing and rightward peak shift. The proximity of the TiC_y diffraction shoulder to the β -SiC (111) reflection further suggests local lattice coherency, facilitating strain transfer across the boundary. This strain state is likely accommodated by the incipient structural ordering and stacking faults discussed in the following sections.

At higher MXene loadings (5 and 10 wt.%), we also observe weak but distinct emergence of graphitic carbon peaks at 25.4° . This is consistent with our in-situ XRD results (Figure 3), which show suppressed β -SiC formation even with its early nucleation. The appearance of this peak suggests that a portion of the carbon in the system crystallizes into a graphitic structure rather than incorporating into the SiC lattice. We hypothesize that the unique layered architecture of the MXene precursor may template this graphitization by geometrically constraining carbon within the interlamellar spaces during pyrolysis, effectively acting as a 'confinement' template. While the current XRD data suggest the presence and systematic scaling of this graphitic phase, indicating the specific confinement mechanism would require spatially resolved Raman microspectroscopy or high-resolution TEM, which is beyond the scope of this study.

Traces of α -SiC are also seen with higher MXene contents. Although α -SiC is thermodynamically stable only above $\sim 1700^\circ\text{C}$, its appearance in the MXene-containing samples at lower temperatures is consistent with

heterogeneous nucleation on MXene-derived interfaces. TiC_y and graphitic-carbon surfaces formed during MXene phase transformation provide crystallographically compatible, low-energy templates that lower the activation barrier and stabilize otherwise subcritical α -SiC nuclei at carbide-carbide and carbide-carbon interfaces.³⁴ In parallel, MXene-driven carbon consumption and redistribution may also disrupt local chemical potentials and diffusion fields, creating microenvironments that favor α -SiC nucleation even while bulk β -SiC growth remains the dominant phase.

Microstructure evolution

Figure 4c,d show the SEM micrographs of the sintered 5 wt.% MXene sample. A pronounced platelet morphology is observed with an average platelet width of $1.51 \pm 0.34 \mu\text{m}$. This morphological evolution is consistent with the presence of the 4H-SiC polymorph independently identified by SAED crystallography (**Figure 5f**). It is well-established that hexagonal α -SiC polytypes exhibit anisotropic growth kinetics,^{35, 36} preferentially extending along the basal plane to form high-aspect-ratio platelets. Consequently, we interpret the emergence of these platelets as the morphological expression of the TiC_y -nucleated 4H-SiC phase. EDS line scans (**Figure 4d, inset**) indicate titanium enrichment localized at these grain boundaries, suggesting that TiC_y serves as the heterogeneous nucleation site. Such chemical segregation is indicative of interface-driven transport or partitioning during consolidation, congruent with the phase-field models of ceramic grain growth in the presence of multiphase additives.³⁷ These Ti-rich interface regions

imply ongoing surface energy reconstruction, an effect seen previously in TEM studies of MXene flakes subjected to controlled elevated temperature annealing,³⁸ where the tendency to reconstruct local order is driven by the minimization of boundary and defect energies. The presence of transition metal carbides in a PDC matrix can also lead to altered interfacial mobility³⁹ or Zener pinning during the 2D to 3D phase transformation stage. Earlier studies on graphene nanoplates have shown such an effect;⁴⁰ however, this is observed for the first time in a 2D MXene-polymer derived ceramic system.

Site-specific TEM lamellae were prepared using a focused ion beam (FIB) microscope to evaluate the interfaces in the composite. TEM micrographs (**Figure 4e,f**) resolve sharply defined interfaces between β -SiC platelets and adjoining TiC_y domains with a structurally coherent interface. The EDS compositional maps (**Figure 4g**) show a spatial segregation of Si and Ti, indicating phase continuity within each region and an absence of detectable solid solutioning or gradient mixing at the micron scale.

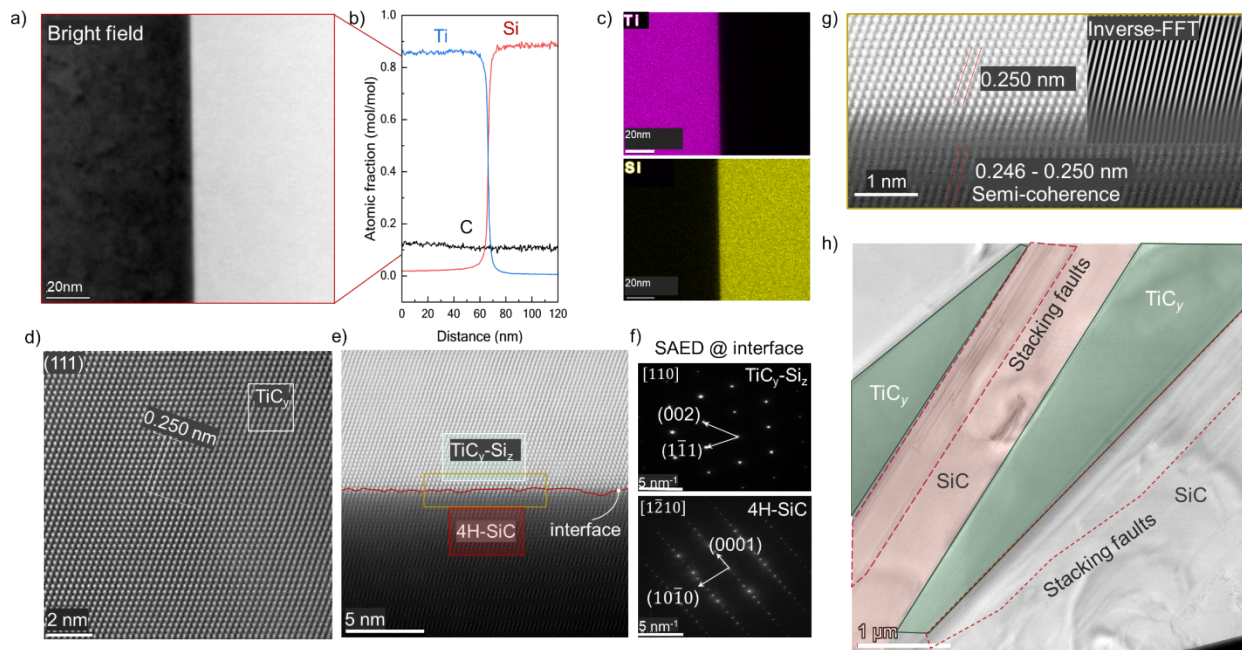


Figure 5: a) High-resolution STEM bright field image of the interfacial region showing b) EDS line spectra of the interface with increasing Ti/Si intensities. The intermixing region is less than 20 nm in width at the junction. c) EDS areal maps showing Ti and Si segregation, d) STEM image of the TiC_y -rich region with a nominal d-spacing of 0.250 nm, e) STEM image of the interface obtained via axial tilt showing 4H-SiC (bottom) and TiC_y -Si region with semi-coherence, f) SAED patterns of the $\text{TiC}_y\text{-Si}_z$ and 4H-SiC at the interface. Coherence is seen along (111) and (0001) planes at the interface. g) Higher magnification STEM image of the interface region showing semi-coherence between the TiC_y and 4H-SiC regions (the inverse FFT inset), h) the TiC-SiC interface micrograph showing the emergence of defects (stacking faults) along the TiC region.

To understand the nature of the interface after establishing the absence of micron-scale solid solution formation, we probed the boundary at higher

spatial resolutions. HAADF-STEM imaging, combined with EDS line scans, exhibit a consistently sharp but structurally coherent interface between SiC platelets and adjacent TiC_y domains. However, when the probe region is restricted to ~ 20 nm across the interfacial region (**Figure 5a,b,c**), subtle Si intermixing becomes apparent within the Ti-rich zone, evolving as a narrow, nanometer-scale transition layer. Quantitative EDS mapping suggests localized Si incorporation at an amount (1.8 ± 0.016 at.%) well below bulk stoichiometry, indicating a partial chemical reconstruction unique to the interface. The estimated stoichiometry based on the EDS data is $\text{Ti}_{31.2 \pm 7.1}\text{Si}_1\text{C}_{5.7 \pm 1.6}$, very Ti-rich. In the Si-rich areas, we observe less than 0.63 ± 0.034 at.% Ti diffusion, again, very little intermixing. Such a stoichiometry is only plausible at interfacial or transition regions, not as a bulk phase.

STEM imaging in the interfacial regions (**Figure 5d,e**) suggests the formation of a TiC_y phase ($0.63 > y > 0.7$) (**Figure 5d**), with measured d-spacing at $\sim 4.327 \pm 0.0042$ Å for the cubic lattice.⁴¹ The SAED patterns acquired from this region exhibits cubic symmetry (**Figure 5f-top, Figure S7**), whereas the immediate interface region shows a hexagonal symmetry with measured in-plane and out-of-plane a/c lattice ratios closer to 4H-SiC (**Table 1**).

Interplanar spacings of ~ 0.250 nm for $\text{TiC}_y(111)$ and ~ 0.246 - 0.250 nm for 4H-SiC (0001) in SAED and inverse FFT analyses (**Figure 5g**) indicate a semi-coherent interface with a well-defined orientation relationship, specifically $(111)\text{TiC}_y \parallel (0001)4\text{H-SiC}$. This relationship arises because the fcc

(111) plane in TiC_y and the hexagonal (0001) basal plane in 4H-SiC are both close-packed layers that share the same 2D hexagonal atomic net; their in-plane lattice parameters are closely matched. In this context, the (0001) notation denotes successive basal planes in the hexagonal stacking sequence (ABAC for 4H-SiC), while (111) denotes the close-packed planes of the fcc TiC_y (ABC stacking);⁴² despite different stacking sequences along the normal, the in-plane atomic packing is topologically equivalent, enabling semi-coherency and efficient strain transfer across the interface. These observations are consistent with our earlier report that MXene-derived cubic carbides preferentially develop along the (111) plane during phase transformation.²⁸ Such an orientation relationship would be expected to lower the interfacial energy for nucleation and facilitate crystallographic alignment of adjacent SiC polytypes. In the present work, we interpret the semi-coherent $(111)\text{TiC}_y \parallel (0001)4\text{H-SiC}$ relationship and the associated stacking faults as signatures of this interfacial accommodation mechanism; however, a fully quantitative description of the interfacial energetics lies beyond the scope of the current study.

Table 1: Indexing of SAED patterns showing deviation from the standard $g(1010)/g(0001)$ ratios.

Sample	$g(1010)/g(0001)$ ratio
4H-SiC	3.84

Ti₃SiC₂	6.64
Ti₂SiC	4.87
This study	3.81

We speculate that the limited intermixing of Si or Ti at the interface regions leads to a transient interface stabilized configuration rich in titanium, which is dictated by local constraints. This shows that while the SiC and TiC_y domains retain their bulk identities on either side of the interface, the junction itself is capable of supporting a distinct phase that mediates chemical and structural coherence across the boundary.

We also observed an increased density of stacking faults proximal to the interfaces (**Figure 5h, Figure S8**), consistent with the evolution of planar defects that relieve misfit and accommodate strain at these semi-coherent TiC_y/SiC boundaries; this interpretation aligns with our earlier XRD interpretations of interfacial ordering and strain (**Figure 4b,c**). In practical terms, the approximately commensurate spacing between fcc-TiC_y (111) and 4H-SiC (0001) planes enables semi-coherency, while the residual in-plane and out-of-plane misfit is preferentially taken up by intrinsic stacking faults that reduce interfacial energy and facilitate slip-assisted relaxation within a few nanometers of the boundary.⁴³ Further studies are required to corroborate these hypotheses.

Bulk solid solution formation within the Ti-Si-C system is constrained both crystallographically and thermodynamically. Structurally, TiC (fcc,

octahedral C sites) and SiC (hexagonal polytypes with tetrahedral coordination) exhibit incompatible site topologies and stacking sequences, limiting substitution across sublattices. Thermodynamically, the phase space is dominated by stable line compounds (e.g., TiC, SiC, Ti_3SiC_2 , TiSi_2 , Ti_5Si_3) characterized by low Gibbs free energies and positive enthalpies of mixing, which further suppress extensive solid solution formation.^{44, 45} However, under non-equilibrium conditions, interfacial transport and localized strain can promote partial elemental partitioning and defect-mediated intergrowths, particularly at grain boundaries and within nanostructured domains.^{46, 47}

A thin Ti-Si-C interfacial layer may form under these conditions, stabilized by strain from lattice mismatch between (111) TiC_y and (0001) 4H-SiC. Out-of-plane mismatch is accommodated by stacking faults and a high concentration of defects introduced by mobile species during consolidation. Given the known propensity of grain interfaces to accommodate defects,^{48, 49} these environments may promote the emergence of locally ordered structures distinct from the bulk MAX phases, yet insufficiently developed to constitute a fully stabilized form. This interfacial nanochemistry enabled through controlled synthesis and in-situ phase evolution can lead to emergent phases inaccessible in the bulk, yet vital for inter-grain coherence in MXene-derived carbides-polymer-derived ceramic composite hybrids. However, given the localized sample volume, overlap of diffracted intensities from adjacent SiC polytypes, and the presence of extensive stacking faults, we cannot unambiguously exclude alternative interpretations such as faulted SiC polytypes and/or mixed contributions from TiC_y variants in this zone. We

interpret the interfacial contrast as evidence of a locally reconstructed Ti-Si-C interphase, rather than as a definitively new bulk crystalline phase.

The current SAED indexing and real-space imaging provide a plausible structural model for this interphase, but a rigorous distinction between new Ti-Si-C compound and defect-stabilized configurations of known phases would require additional atomic-resolution spectroscopy and first-principles modeling.

As the field of 2D MXenes continues to expand, this work provides a foundation for leveraging their unique microstructure from their derived carbides for high-performance ceramic systems. The demonstrated synthesis-structure relationships not only advance our understanding of interface-mediated phenomena in hybrid 2D-carbide and PDC composites but also establish design principles for the rational development of MXenes as additives in PDCs.

Conclusions

This study focuses on understanding 2D $\text{Ti}_3\text{C}_2\text{T}_x$ MXene-derived carbide-SiC composites. 2 wt.% MXene loading yields thermal stability and reduces mass loss by 2.45-4.21%, a concentration threshold where MXene sheets effectively stabilize the evolving ceramic network without introducing detrimental mass transport limitations. In situ high-temperature XRD shows earlier crystallization with MXene-containing composites, achieving crystalline β -SiC formation at temperatures 80°C lower than the pure SiC system. However, MXene also shows suppression of SiC grain coarsening. We

identify a nanometer-scale Ti-rich Ti-Si-C interphase at $\text{TiC}_y\text{-SiC}$ boundaries, exhibiting hexagonal-like stacking and a semi-coherent orientation relationship to 4H-SiC. This locally reconstructed interfacial region, together with the observed stacking-fault density, provides new structural insight into how MXene-derived carbides interact with polymer derived SiC grains during high-temperature processing. Although these features plausibly influence intergranular coherence and crack-path selection, their mechanical consequences have not yet been quantified and will be the focus of future property-driven studies. These transient, strain-stabilized interphases, confined to ~ 20 nm regions at $\text{TiC}_y\text{-SiC}$ boundaries, appear as metastable configurations that are not accessible as bulk phases but may play an important role in mediating chemical and structural integrity across the interface.

End of Document Items

CRediT Authorship Statement

- Srinivasa Kartik Nemani: Methodology, Conceptualization, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - Original draft, Writing - Review & editing, Visualization.
- Kaustubh K. Bawane: Investigation, Resources, Data curation, Writing - Review & editing, Visualization.
- Brian C. Wyatt: Methodology, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - Review & editing.
- Mubina Shaik: Investigation, Resources.

- Apurba Naskar: Resources.
- Babak Anasori: Methodology, Resources.
- Kathy Lu: Conceptualization, Resources, Writing - Review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data required to reproduce these findings are available upon reasonable request.

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